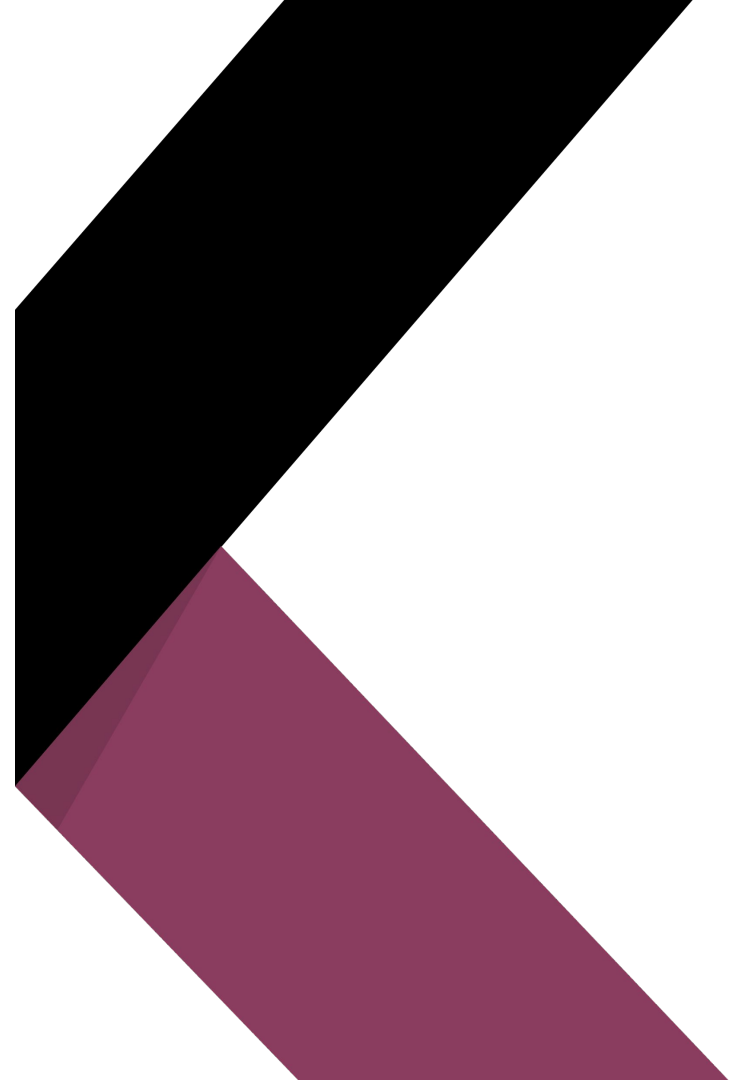


Academy

Ethernet





Why is Ethernet the revolution?

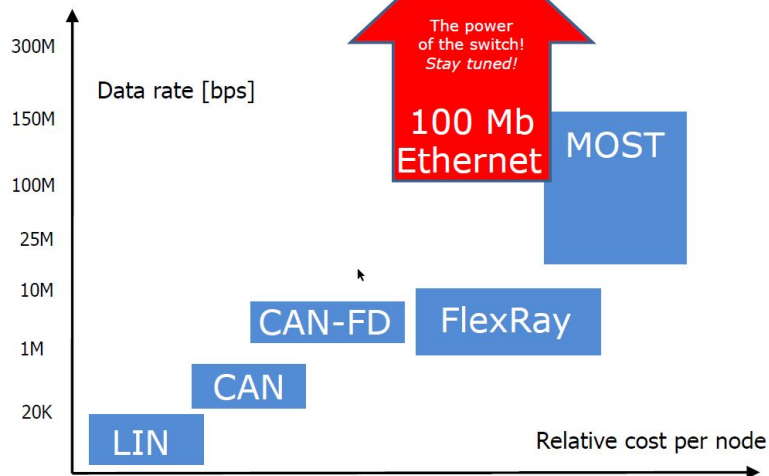
MOST COMMON TECHNOLOGY

1. LIN
2. CAN, CAN-FD
3. FlexRay
4. MOST

Ethernet is the revolution

1. Bandwidth
2. The Ethernet is the technology used by the majority of devices

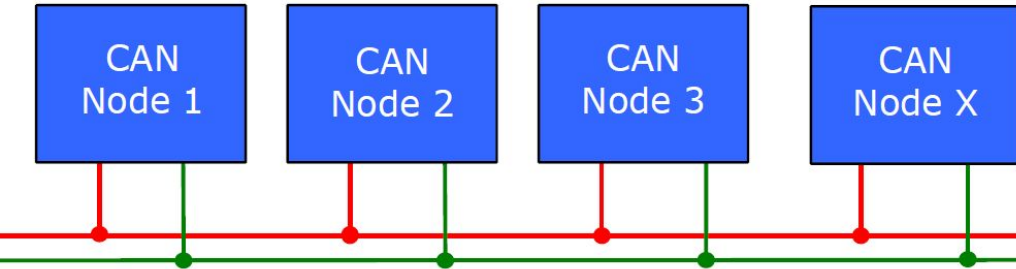
Cost vs. Data Rate





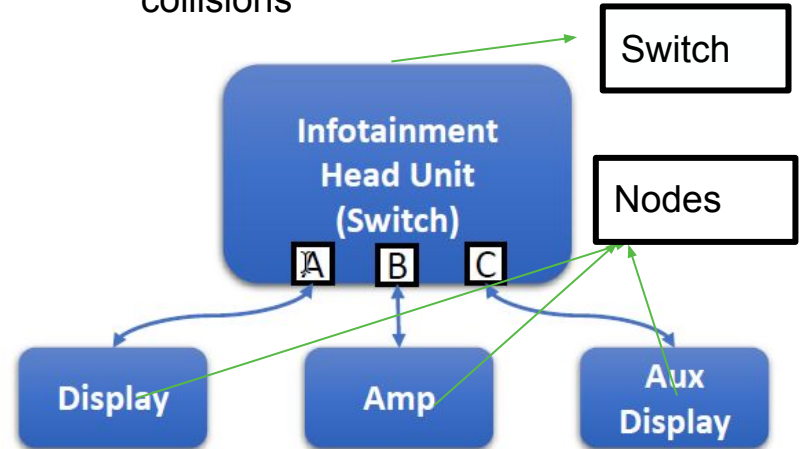
CAN vs Ethernet

Comparison



1. Switch-Nodes (ONE-Many configuration)
2. Up to 1Gb/s for standard CAN
3. Single twisted pair copper wire
4. Plug and play
5. communication is signal oriented
6. Arbitration based method to handle collisions

1. MultiDrop Topology with the ability to add and remove nodes
2. Up to 1Mb/s for standard CAN
3. Single twisted pair copper wire
4. Plug and play
5. communication is signal oriented
6. Arbitration based method to handle collisions

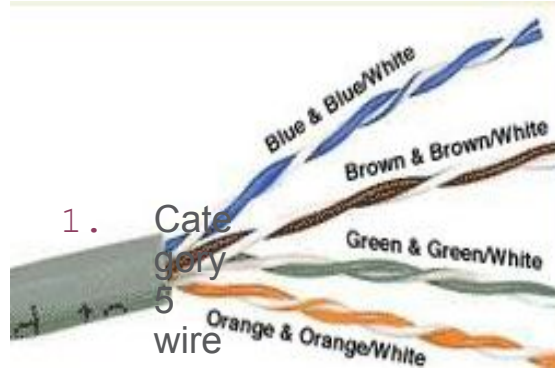




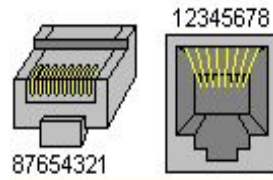
Bandwidth Different Technology

Ethernet wire

1. Ethernet cable at least Category 5 (Cat5)
2. Inside the ethernet cable, there are 8 color coded wires. These wires are twisted into 4 pairs of wires, each pair has a common color theme.

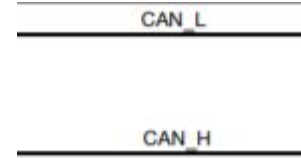


1. Here is a wiring diagram and pin out:
2. Modular Connector Plug and Jack Pin Out



CAN wire

1. Inside the CAN cable, there are 2 wires. CAN_L and CAN_H the information is the difference of the voltage



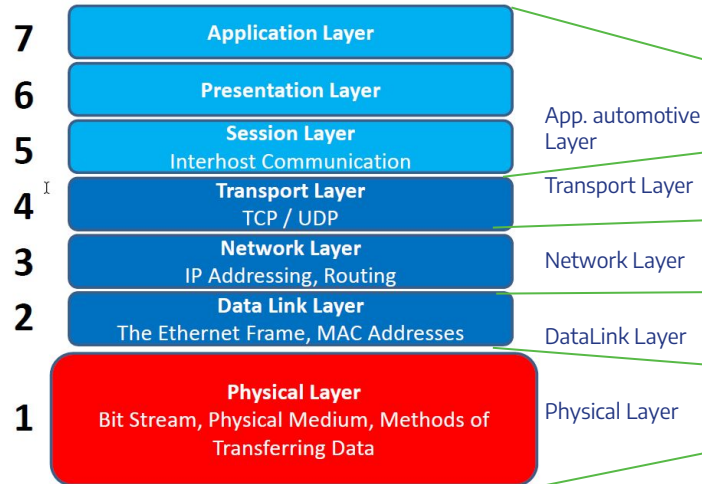


Encapsulation

Model

General model of ethernet network

All the layers are described in detail

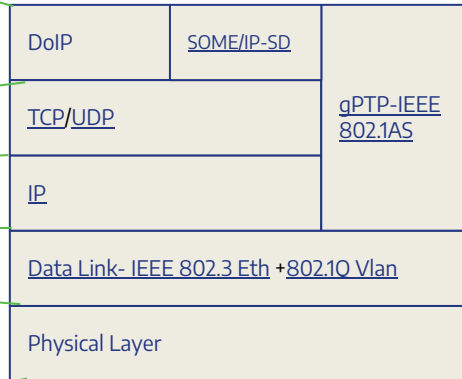


Model of automotive Ethernet

The application, presentation and session layers are compressed in App. automotive layer

Model of CAN

The application, presentation and session layers are compressed in App. CAN layer



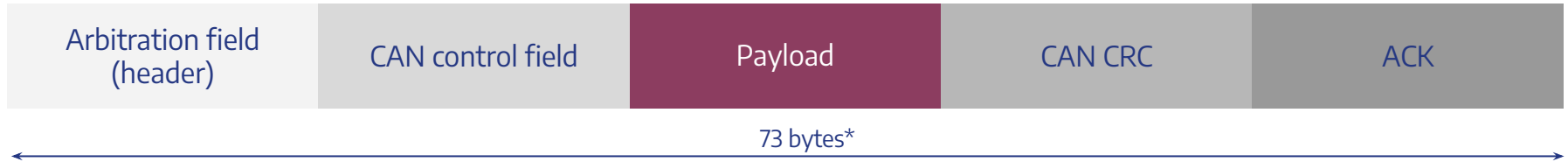


Encapsulation

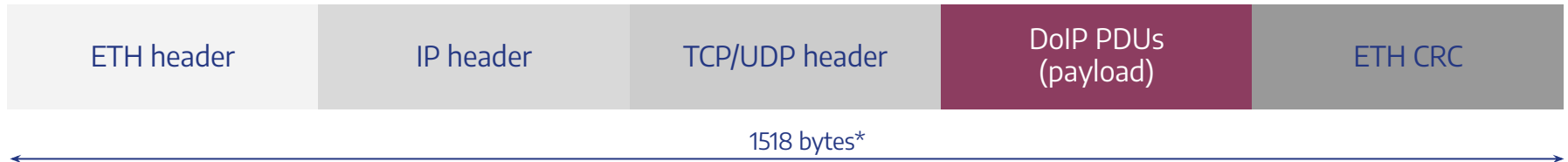
Frame

1. Length of the message
2. Number of the fields

CAN FD frame



ETH frame



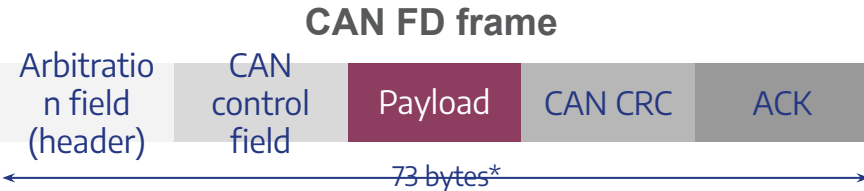


Service oriented architecture



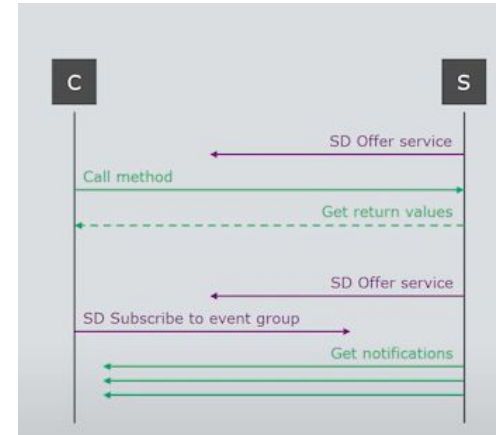
Architecture comparison

CAN technology is signal oriented. In other words we have a message frame each field has a fixed bits length with a specific meaning



The main information is brought by payload

Ethernet application technology is service oriented. In other words we send/require a service. It can trigger a lot of messages.



The S (Server) offers some services to C (Client). C responds OK. Now the server can send the information



1. SOME/IP provides service oriented communication over a network. It is based on service definitions that list the functionality that the service provides. The SOME/IP communication protocols allow the exchange of complex data types in various ways such as remote procedure calls (RPC), Notification events and fields
2. • SOME/IP offers a wide range of middleware features like **serialization, Remote Procedure Call (RPC), Service Discovery, Publish/Subscribe** to enable the ECU software to communicate with each other.

Serialization

It is the way the data is represented in a data unit, which can be either a UDP or a TCP message. Interoperability between different ECU can be ensured only if there is a mechanism for a consistent data transmission. SOME/IP allows for some serialization.

Remote Procedure

the client can ask for data as a response or simply call a function to perform some task at the server-end.

Service discovery

A client knows which services are currently available, SOME/IP-SD provides two mechanisms that allow the dynamic discovery of services. An Offer Service enables the server to provide the network with all its offered services. The Find Service allows clients to request available services instead.

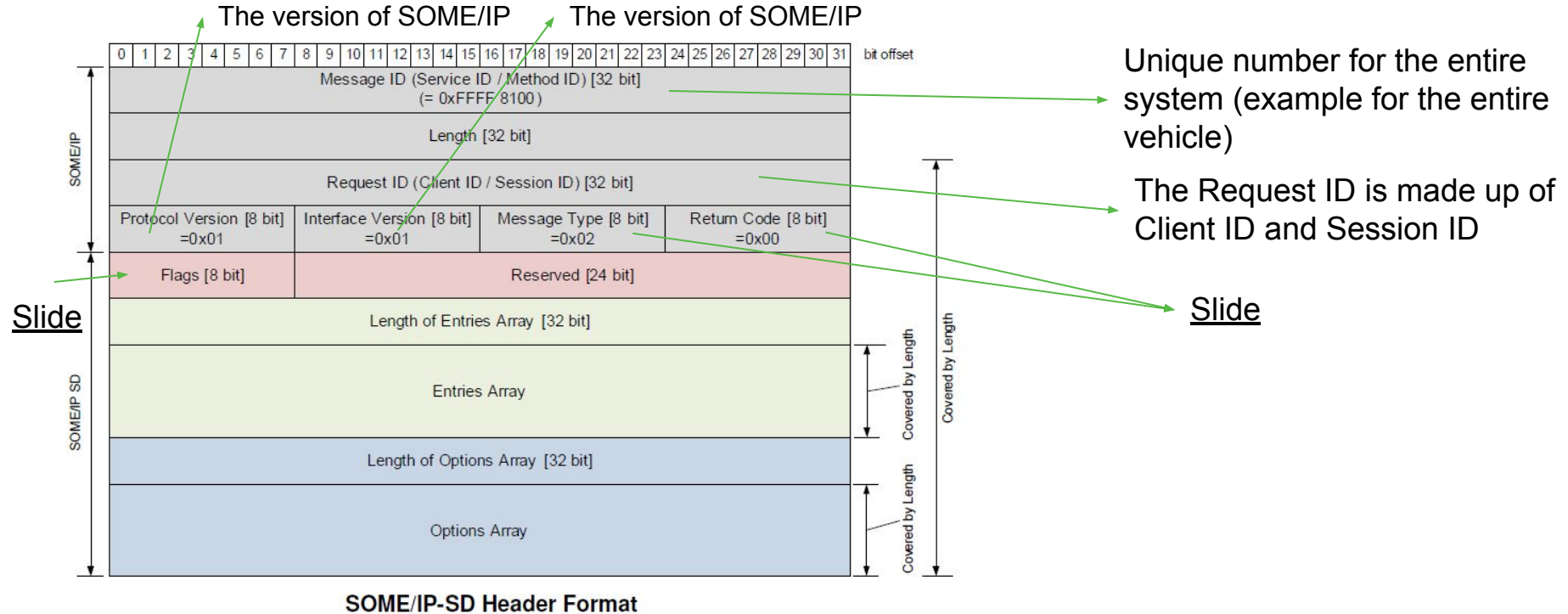


SOME/IP-SD

Message frame

SOME/IP-SD is used to

- Locate service instances.
- Detect if service instances are running.
- Implement the Publish/Subscribe handling.

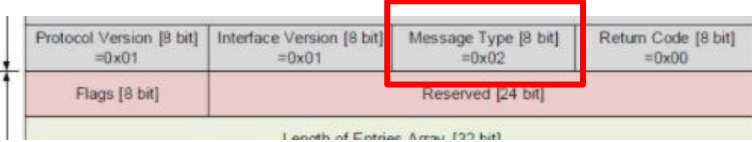




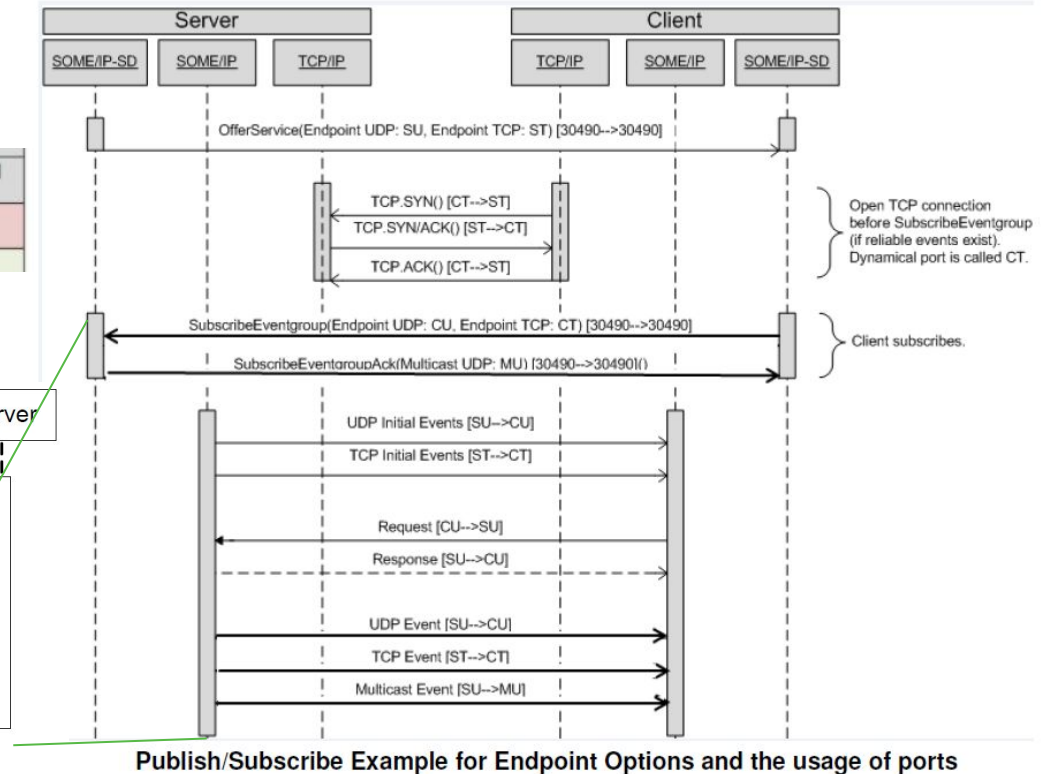
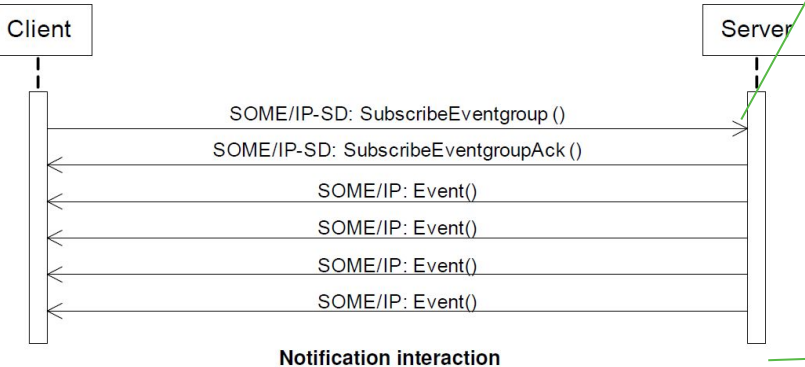
Example

Publish/Subscribe communication

The application layer SOME/IP-SD offer a service:



Type=**0x02** (Notification) SD use only
Notification message.

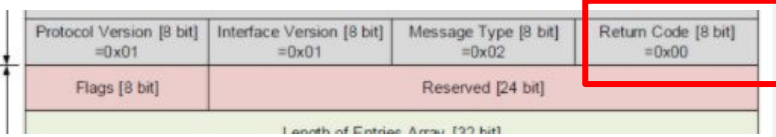




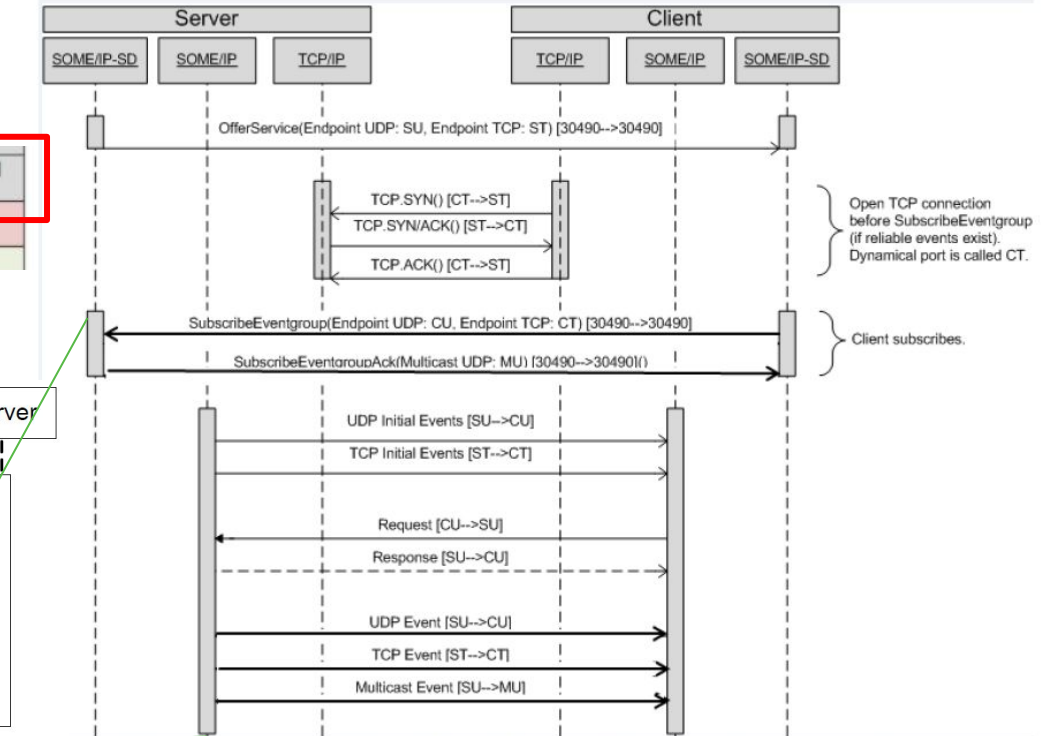
Example

Publish/Subscribe communication

The application layer SOME/IP-SD offer a service:



Return Code=0x00 (Notification) SD do not acknowledge the right process of the return message.



Notification interaction

Publish/Subscribe Example for Endpoint Options and the usage of ports

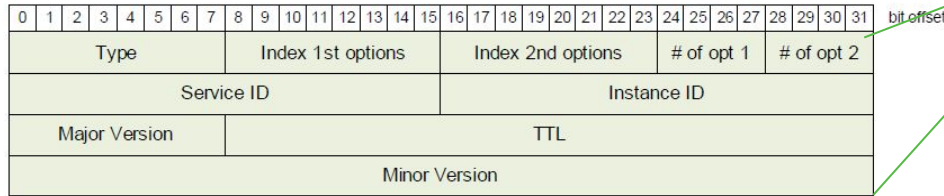


Example

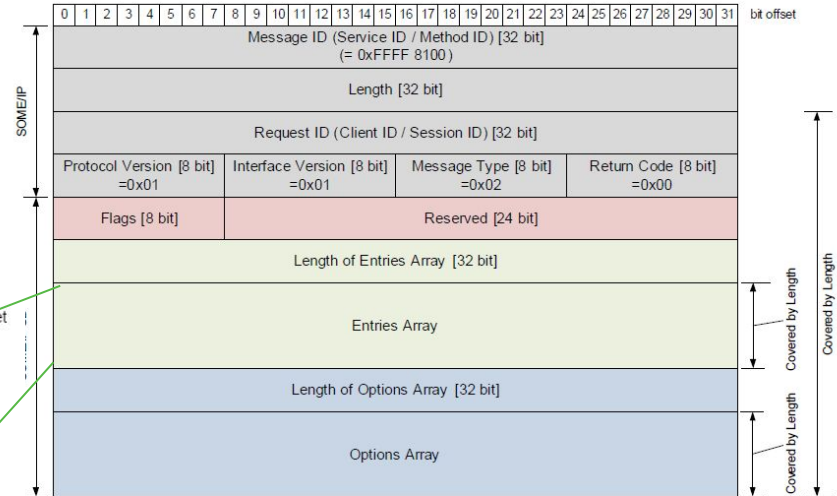
Publish/Subscribe communication

The subscription information are carried by the entries. Exist two types of entries:

1. Entries for Services
2. Entries for Eventgroups



SOME/IP-SD Service Entry Type



SOME/IP-SD Header Format

Type Field [8 bits]

The Type field of the Service Entry format layout shall carry one of the following values:

- 0x00 to encode FindService
- **0x01 to encode OfferService** and StopOfferService.



Example

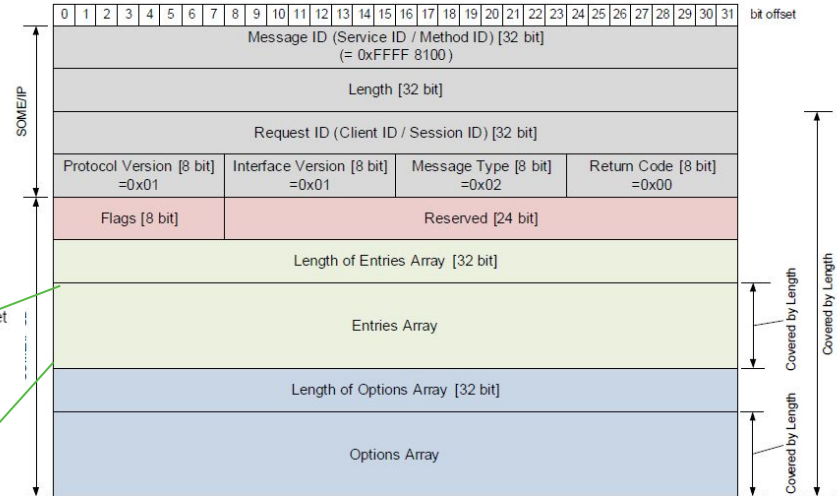
Publish/Subscribe communication

The application layer SOME/IP-SD stops to offer a service:

1. (TCP/IP frame)TTL= 0x000000.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	bit offset
Type								Index 1st options								Index 2nd options								# of opt 1				# of opt 2				
Service ID																Instance ID																
Major Version								TTL																								
Minor Version																																

SOME/IP-SD Service Entry Type



SOME/IP-SD Header Format

Service ID field[16 bits]

Describes the Service ID of the Service or Service-Instance this entry is concerned with.

Instance ID[16 bits]

Describes the Service Instance ID of the Service Instance this entry is concerned with or is set to 0xFFFF if all service instances of a service are meant.



Transport layer



TCP/IP

Transport layer

App. automotive
Layer

Transport Layer

Network Layer

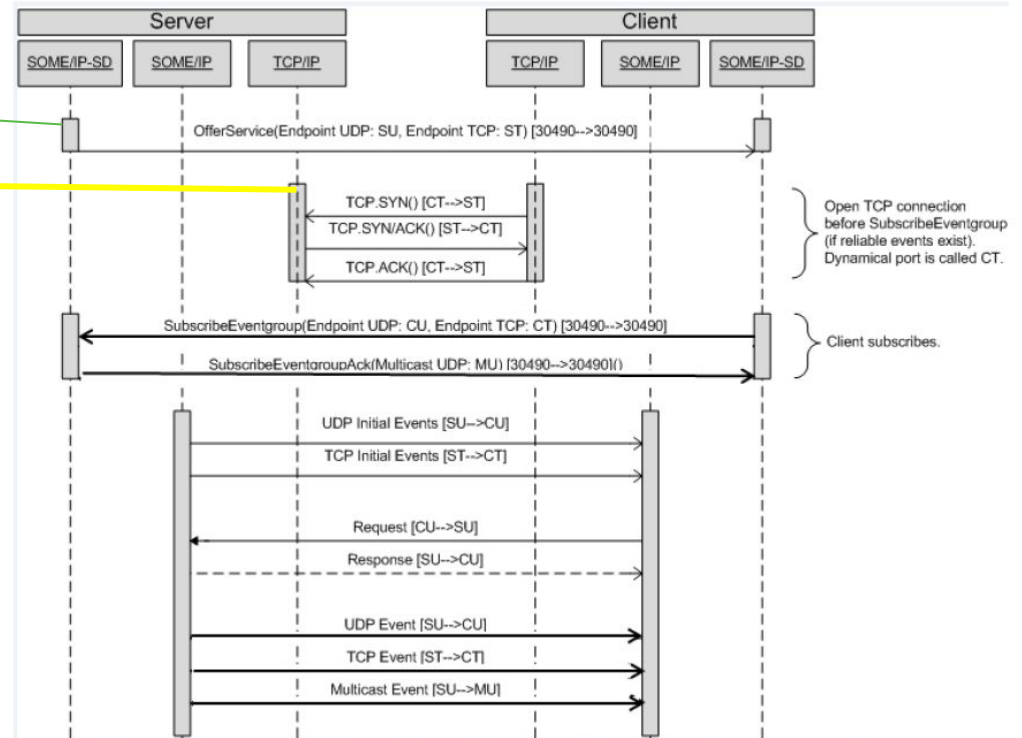
DataLink Layer

Physical Layer

DoIP	<u>SOME/IP-SD</u>	<u>gPTP-IEEE 802.1AS</u>
<u>TCP/UDP</u>		
<u>IP</u>		
<u>Data Link- IEEE 802.3 Eth +802.1Q Vlan</u>		
Physical Layer		

The SOME/IP-SD services can be provided thanks to the lower layers:

- Transport layer
- Network layer
- DataLink Layer
- Physical Layer



Publish/Subscribe Example for Endpoint Options and the usage of ports

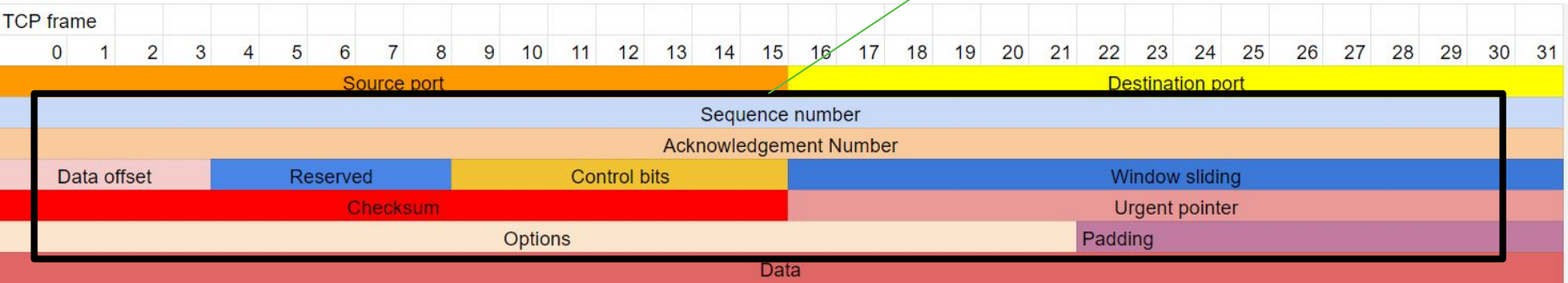


Message frame

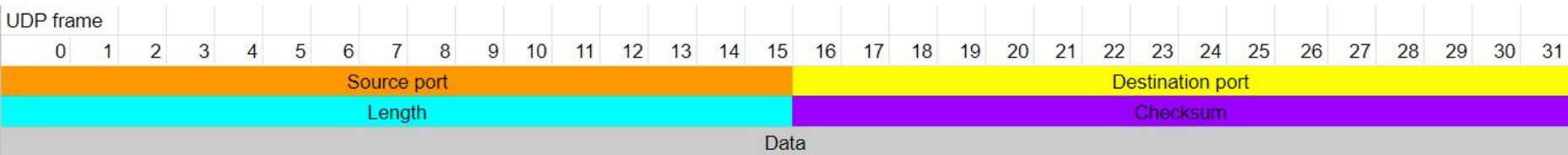
Transport protocol

The main difference between TCP and UDP is the highlighted fields(deltail)

TCP frame



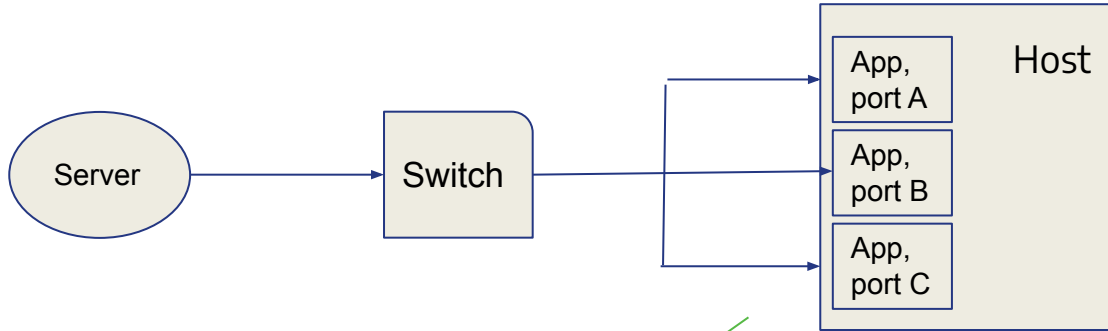
UDP frame





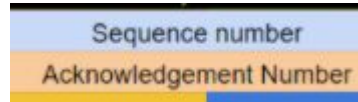
Common features

transport protocol



TCP frame

1. TCP protocol is connection-oriented, and a connection between client and server is established before data can be sent
2. TCP detects these problems, requests re-transmission of lost data, rearranges out-of-order data and even helps minimize network congestion to reduce the occurrence of the other problems



Virtual application addressing:

How to send a message to a specific host and specific application



UDP frame

UDP frame is simpler and shorter than TCP frame so it is more faster in the communication

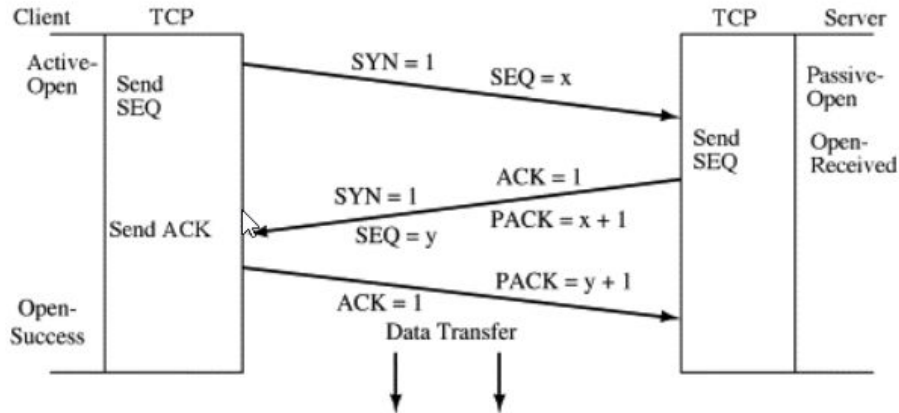


TCP frame

Connection oriented

Purpose

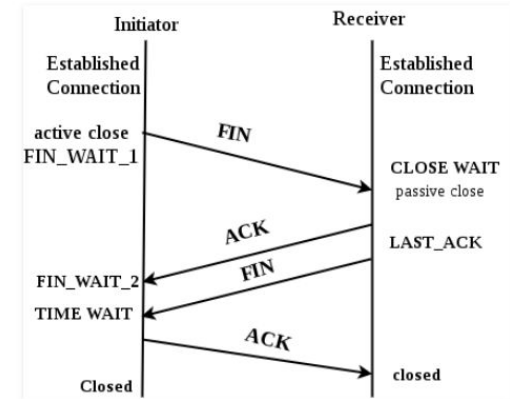
- Connection establishment
- Ordering of datagrams



After the established connection the server sends X bytes. The seq number is 1 because it points the first byte.
The next message of client is an ACK (the client says to client that want the next byte).
The termination of connection is set with FIN, ACK

Main elements: Client and Server connection established

- The client start a connection with **SYN message** the server knows now the seq number of client
- The server respond with **SYN-ACK message** the server request a seq number of sender with ack number= seq num+1
- The client respond with an ACK message and they establish a connection





Network layer



IP layer

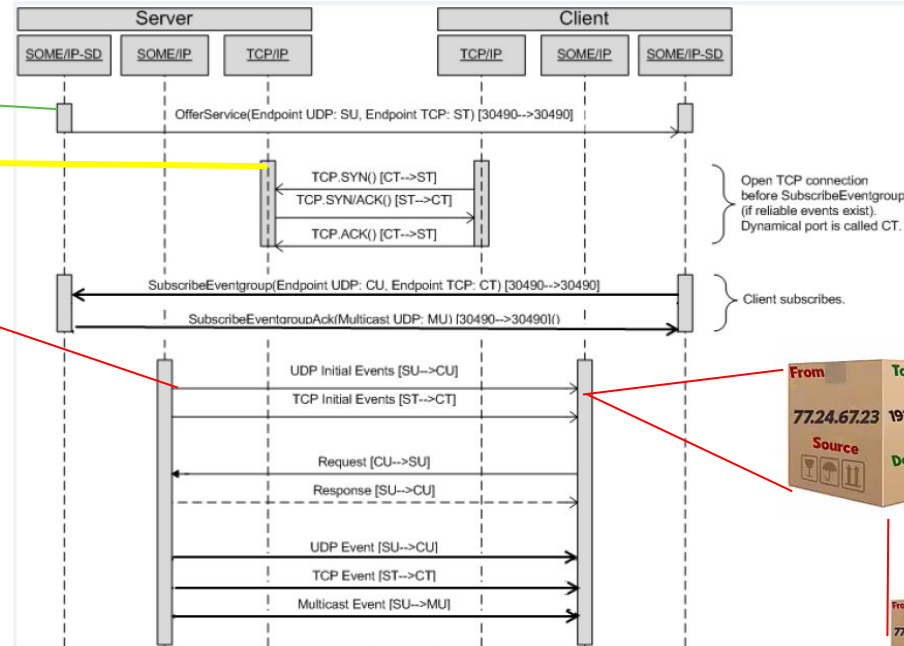
Network layer

Main features:

1. Route the packets
2. Segmentation of packets

App.
automotive
Layer
Transport
Layer
Network
Layer
DataLink
Layer
Physical Layer

DoIP	<u>SOME/IP-SD</u>	<u>gPTP-IEE E 802.1AS</u>
<u>TCP/UDP</u>		
<u>IP</u>		
<u>Data Link- IEEE 802.3 Eth +802.1Q Vlan</u>		
Physical Layer		



Publish/Subscribe Example for Endpoint Options and the usage of ports

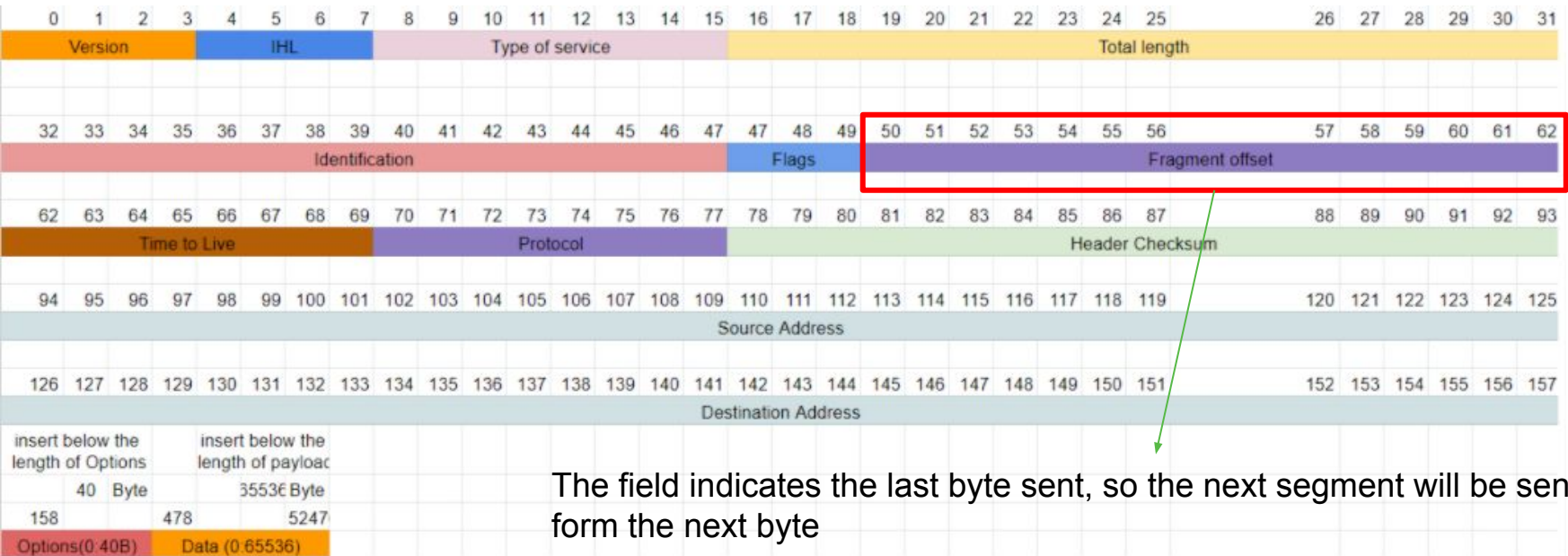




message frame



is described by the following message frame. If the packet is too large and the bit 2 of the flag is set 1. It can be segmented into different segments



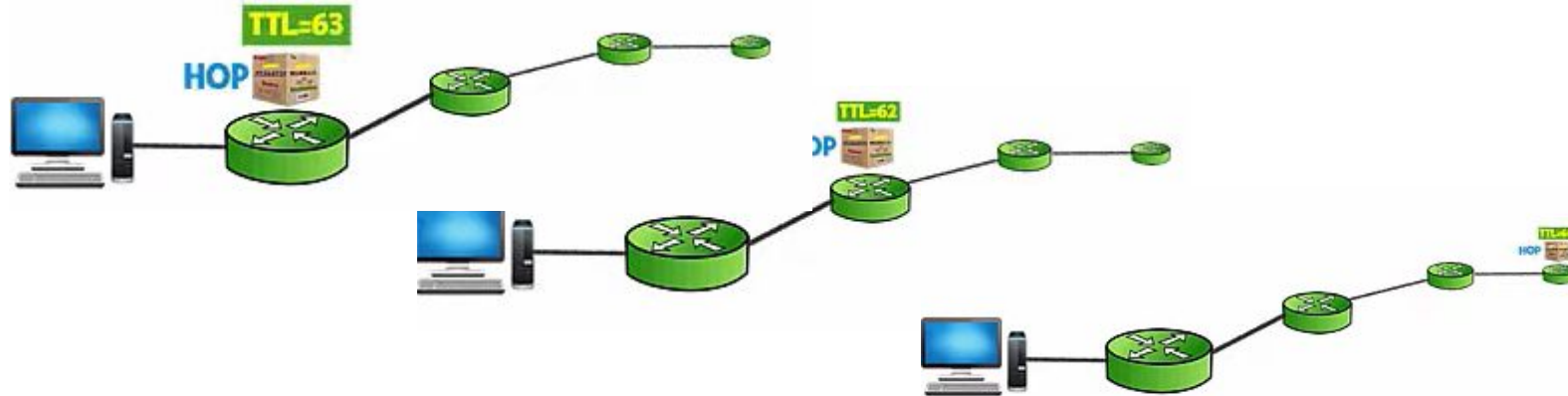


TTL

field

TTL time to live

TTL value starts with the sender, generally the initial value is 64 each router decrease of 1
Each passage of a packet from a node is designated a hop

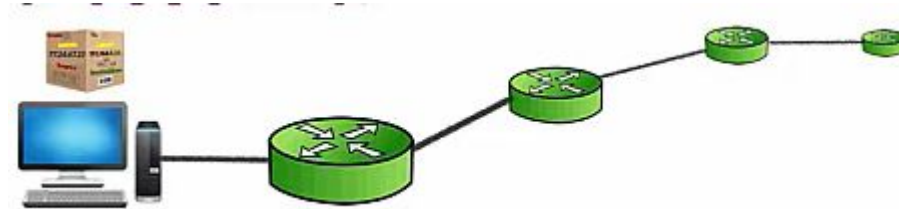




Checksum Field

Header checksum

This field ensure the integrity of the header field values. It is applied on the IP packet header. The field itself is not involved in the calculation assuming it the value 0 (in the calculation)



The check of the integrity of the error is essential because the packets pass through a lot of routers that modify the header of the packet so this field is very important



Classful Networks- How does it work the ip addressing?

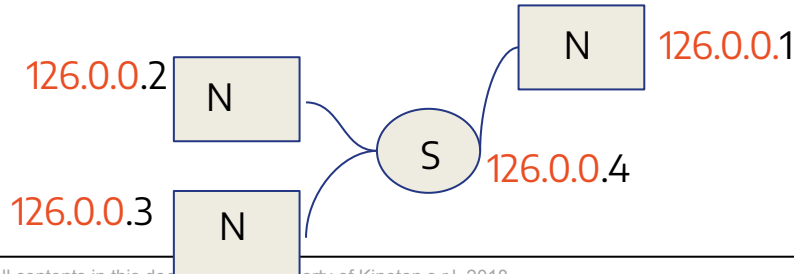
Each ip address is made by:

1. Network
2. Host

The method divides the IP address space for Internet Protocol version 4 (IPv4) into five address classes [A,B,C,D,E] based on the leading four address bits.

Classes A, B, and C provide unicast addresses for networks of three different network sizes. Class D is for multicast networking and the class E address range is reserved for future or experimental purposes

Ex Class A Subnet mask 255.0.0.0, it means the \8 MSBs (most significant bits are fixed to set the network) the other bits (32-8) are set to define the nodes of the network



	Start address	End address	Default subnet mask in dot-decimal notation	CIDR notation
A	0.0.0.0	127.255.255.255 ^[a]	255.0.0.0	/8
B	128.0.0.0	191.255.255.255	255.255.0.0	/16
C	192.0.0.0	223.255.255.255	255.255.255.0	/24
D	224.0.0.0	239.255.255.255	not defined	not defined
E	240.0.0.0	255.255.255.255 ^[b]	not defined	not defined



Ethernet layer



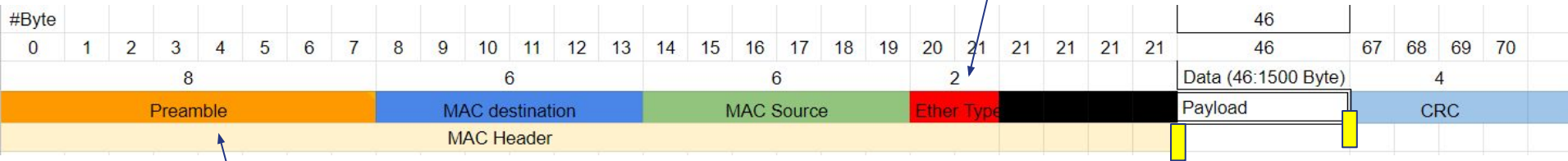
Ethernet

frame

The format of the eth frame changes depending on Ether Type:

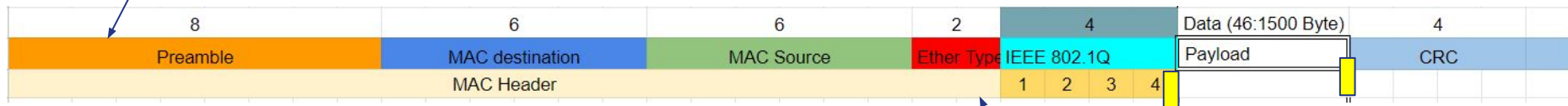
Ethernet I

Indicate the structure of the upper layer



Ethernet II

A series of Bytes (1 and 0 alternatively), they indicate a frame is coming, and enables synchronization of the transmitter and receiver

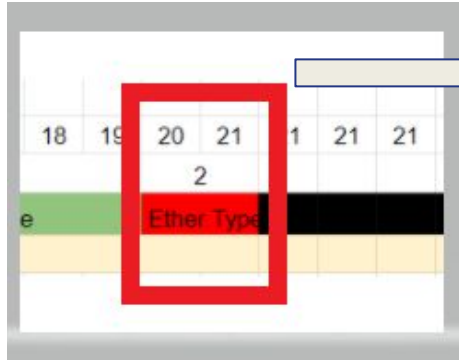


0x8100



EtherType Ethernet frame

The value of the EtherType set the format of the following fields



The **main values** of the field
Ether Type

0x86DD	Indicate a datagram IPv6
0x0806	Indicate a frame ARP
0x0800	Indicate a datagram IPv4
0x8100	indicate a frame IEEE 802.1Q



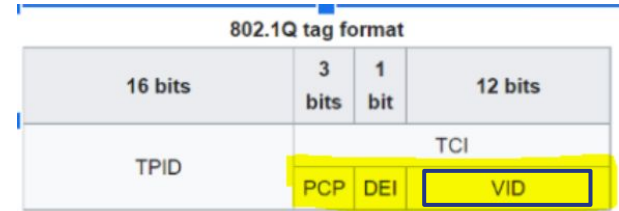
Format of the Q802.1Q tag

TPID Tag protocol identifier

PCP Set the priority of the frame

DEI Dropping packets in case of congestion

VID Indicate the ID of Vlan



Example of TCI field

0x000 → no VLAN

0xFFF → reserved



no VLAN → The packet belongs to default VLAN
Reserved → This value is not assigned to specific feature



Details



XX

XX

Message Type [8 bit]

The Message Type field is used to differentiate different types of messages:

Number	Value	Description
0x00	REQUEST	A request expecting a response (even void)
0x01	REQUEST_NO_RETURN	A fire&forget request
0x02	NOTIFICATION	A request of a notification/event callback expecting no response
0x80	RESPONSE	The response message
0x81	ERROR	The response containing an error)
0x20	TP_REQUEST	A TP request expecting a response (even void)
0x21	TP_REQUEST_NO_RETURN	A TP fire&forget request
0x22	TP_NOTIFICATION	A TP request of a notification/event call-back expecting no response
0x23	TP_RESPONSE	The TP response message
0x24	TP_ERROR	The TP response containing an error)

Return Code [8 bit]

The Return Code shall be used to signal whether a request was successfully processed. For simplification of the header layout, every message transports the field Return Code.

Message Type	Allowed Return Codes
REQUEST	N/A set to 0x00 (E_OK)
REQUEST_NO_RETURN	N/A set to 0x00 (E_OK)
NOTIFICATION	N/A set to 0x00 (E_OK)
RESPONSE	See Return Codes in [PRS_SOMEIP_00191]
ERROR	See Return Codes in [PRS_SOMEIP_00191]. Shall not be 0x00 (E_OK).

Allowed Return Codes for specific Message Types



Flags

8 bit

Flags field [8 bits]

With the Flags field the SOME/IP-SD header starts. It is used to signal global Service Discovery information, which includes currently the state of the last reboot as well as the capability of receiving unicast messages.

Reboot Flag[1 bit]

The Reboot Flag of the SOME/IP-SD Header shall be set to one for all messages after reboot until the Session-ID in the SOME/IP-Header wraps around and thus starts with 1 again. After this wrap around the Reboot Flag is set to 0. The information for the reboot flag and the Session ID shall be kept for multicast and unicast separately.

Unicast Flag[1 bit]

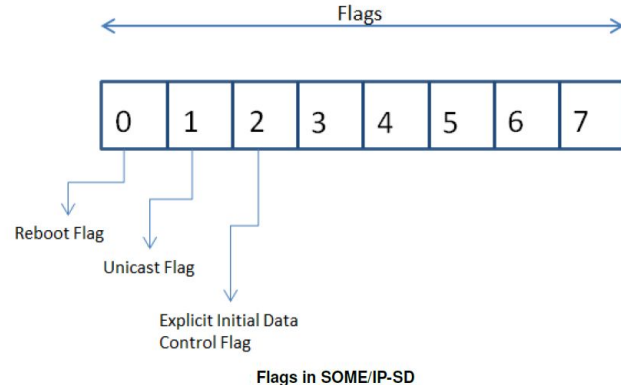
The second flag of the SOME/IP-SD Flags (second highest order bit) shall be called Unicast Flag. The Unicast Flag of the Flag field shall be set to Unicast Flag and shall be set to '1', meaning: This ECU supports receiving Unicast messages.

Explicit Initial Data control Flag[1 bit]

The third flag of the SOME/IP-SD Flags (third highest order bit) shall be called Explicit Initial Data Control Flag and shall mean that the ECU supports explicit initial data control. This means that the ECU understands and honors the Initial Data Requested Flag inside of Eventgroup Entries.

Other Flag bits[5 bits]

Undefined bits within the Flag field shall be statically set to '0'.





TCP

fields

Source port destination port

Source port and Destination port are 16 bit field

Sequence number

32 bit field if SYN is present the seq number is the ISN (initial sequence number +1)--> essential in the rearrangement of packets

ACK Number

If ACK flag is set, the field contain the next sequence number

Data Offset

The number of 32 bit words in the TCP Header. It indicate where the data begins.

Reserved

6 bit Must be zero

Control Bits

URG: Urgent Pointer field significant
ACK: Acknowledgment field significant
PSH: Push Function
RST: Reset the connection
SYN: Synchronize sequence numbers
FIN: No more data from sender

Windows slide

16bit indicate the buffer of receiving of the receiver

Urgent pointer

16 bit this field is an offset from the sequence number indicating the last urgent databyte

Checksum

Go to [slide](#)

Urgent pointer

Set to 1 (control bit) the pointer contains the position where the data ends

Options

Refer to [link](#)

Data

It is variable field, the length depends on the payload

CONTACTS

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